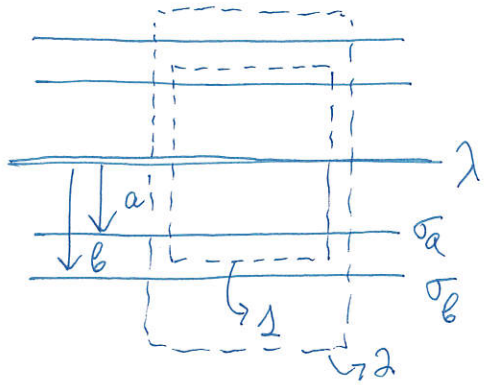
P1



1) let first Gaussian surface be a cylinder of radius $a < r < b$, so that it passing inside conducting shell. Electric field inside conductor ~~is~~ zero

$$\Rightarrow \oint_{\text{surface 1}} \vec{E} \cdot d\vec{A} = 0 = \frac{1}{\varepsilon_0}\left(\lambda L + 2\pi a L \sigma_a\right)$$

charge on the line — charge on inner surface

$$\Rightarrow \sigma_a = -\frac{\lambda}{2\pi a}.$$

2) The second Gaussian surface has $r > b$

$$\Rightarrow \oint_{\text{surface 2}} \vec{E} \cdot d\vec{A} = E(r) \cdot 2\pi r L = \frac{1}{\varepsilon_0} L\left(\lambda + 2\pi a \sigma_a + 2\pi b \sigma_b\right)$$

but the cylindricall shell is not charged $\Rightarrow 2\pi a \sigma_a + 2\pi b \sigma_b = 0$

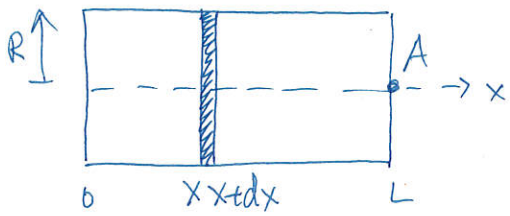
$$\sigma_b = -\frac{2\pi a \sigma_a}{2\pi b} = \frac{\lambda}{2\pi b}.$$

Not required

Interestingly, $E(r) = \dfrac{\lambda}{2\pi\varepsilon_0 r}$

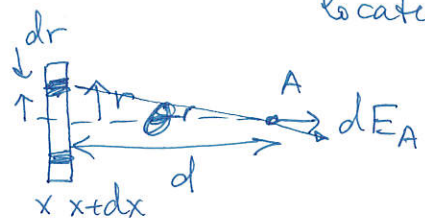
just as in a case of single charged line

P2



By symmetry, $\vec{E}_A = E_A \hat{x}$
We find E_A by adding contributions from disks located at x and of width dx.

Consider single disk and find dE_A by summing over circular shells:



$$\cos\theta = \frac{d}{\sqrt{r^2+d^2}}$$

$$d = L - x$$

$$dE_A = \frac{1}{4\pi\varepsilon_0}\int \frac{dq}{r^2+d^2}\cos\theta$$

$$dq = \rho\, dx\, 2\pi r\, dr$$

$$dE_A = \frac{\rho\cdot 2\pi dx}{4\pi\varepsilon_0}\int_0^R dr\, r\, \frac{d}{(r^2+d^2)^{3/2}} = \frac{\rho\, dx\, d}{4\varepsilon_0}\int_0^{R^2}\frac{du}{(u+d^2)^{3/2}}$$

$(u = r^2)$

$$= \frac{d\rho\, dx}{4\varepsilon_0}\left(-2\frac{1}{\sqrt{u+d^2}}\right)\Bigg|_0^{R^2} = \frac{\rho\, dx}{2\varepsilon_0}\, d\left(\frac{1}{d} - \frac{1}{\sqrt{d^2+R^2}}\right)$$

$$= \frac{\rho\, dx}{2\varepsilon_0}\left(1 - \frac{d}{\sqrt{R^2+d^2}}\right) \quad \text{for } d>0.$$

This is essentially P2.6 of Griffiths

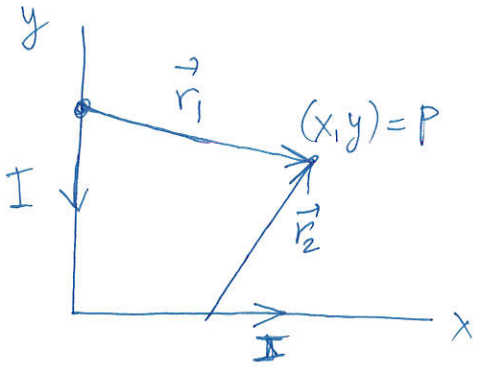
For the cylinder $E_A = \frac{\rho}{2\varepsilon_0}\int_0^L dx\left(1 - \frac{x}{\sqrt{R^2+x^2}}\right) =$

$$= \frac{\rho}{2\varepsilon_0}\left(L - (\sqrt{L^2+R^2} - R)\right) = \frac{\rho}{2\varepsilon_0}\left(L + R - \sqrt{L^2+R^2}\right)$$

Limit: $R \gg L \Rightarrow E_A \approx \frac{\rho}{2\varepsilon_0}(L+R-R) = \frac{\rho L}{2\varepsilon_0}$ ← el. field of a plane with surface charge density ρL.

$L \gg R \Rightarrow E_A = \frac{\rho R}{2\varepsilon_0}$

P3



$$\vec{B}_P = \underbrace{\frac{\mu_0 I}{4\pi}\int_{\text{wire along } y} \frac{d\vec{l}' \times \hat{r}_1}{r_1^2}}_{\vec{B}_y} + \underbrace{\frac{\mu_0 I}{4\pi}\int_{\text{wire along } x} \frac{d\vec{l}' \times \hat{r}_2}{r_2^2}}_{\text{denote } \vec{B}_x}$$

$$d\vec{l}' = -dy'\,\hat{y}$$

$$\vec{r}_1 = x\,\hat{x} + (y-y')\,\hat{y}$$

$$d\vec{l}' \times \vec{r}_1 = x(-dy')\,\hat{y}\times\hat{x} = x\,dy'\,\hat{z}$$

$$\vec{B}_y = \frac{\mu_0 I}{4\pi}\int_0^{\infty} \frac{x\,dy'\,\hat{z}}{\left[x^2+(y-y')^2\right]^{3/2}} = \frac{\mu_0 I\,\hat{z}}{4\pi}\int_y^{-\infty} \frac{-du}{(x^2+u^2)^{3/2}}$$

$u = y-y'$
$du = -dy'$

$$= +\frac{\mu_0 I\,\hat{z}\,x}{4\pi}\int_{-\infty}^{y} \frac{du}{(x^2+u^2)^{3/2}} = \frac{\mu_0 I\,x\,(\hat{z})}{4\pi}\,\frac{1}{x^2}\left(\frac{y}{\sqrt{y^2+x^2}} - \frac{(-L)}{\sqrt{L^2+x^2}}\right) = (L=\infty)$$

Use $\displaystyle\int \frac{du}{(u^2+x^2)^{3/2}} = \frac{1}{x^2}\,\frac{u}{\sqrt{u^2+x^2}}$

$$= \frac{\mu_0 I\,\hat{z}}{4\pi}\,\frac{1}{x}\left(1+\frac{y}{\sqrt{y^2+x^2}}\right)$$

$\vec{B}_x$: $d\vec{l}' = dx'\,\hat{x}$

$$\vec{r}_2 = (x-x')\,\hat{x} + y\,\hat{y}$$

$$d\vec{l}' \times \vec{r}_2 = dx'\,y\;\hat{x}\times\hat{y} = y\,dx'\,\hat{z}$$

$$\vec{B}_x = \frac{\mu_0 I \hat{z}}{4\pi}\int_0^\infty \frac{y\,dx'}{\left[(x-x')^2+y^2\right]^{3/2}} = (u = x-x')$$

$$= \frac{\mu_0 I \hat{z}}{4\pi}\, y \int_{-\infty}^{x} \frac{du}{(u^2+y^2)^{3/2}} = \frac{\mu_0 I \hat{z}}{4\pi}\, y\;\frac{1}{y^2}\,\frac{u}{\sqrt{u^2+y^2}}\Bigg|_{-\infty}^{x}$$

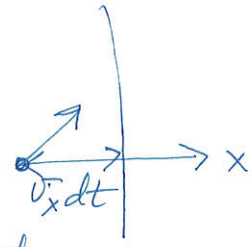
$$= \frac{\mu_0 I \hat{z}}{4\pi}\;\frac{1}{y}\left(\frac{x}{\sqrt{x^2+y^2}} - (-1)\right)$$

$$\vec{B}_p = \frac{\mu_0 I \hat{z}}{4\pi}\left(\frac{1}{x} + \frac{y}{x\sqrt{y^2+x^2}} + \frac{1}{y} + \frac{x}{y\sqrt{x^2+y^2}}\right).$$

The same result can be obtained by following Example 5.5 in Griffiths's book (page 217 and Fig. 5.19)

P4 Number of collisions with area A, per unit time?

$$(v_x dt) f(\vec{p})\, dp_x\, dp_y\, dp_z$$



Particle, with $\vec{p} = (p_x, p_y, p_z)$, which is at distance $v_x dt$ from the wall, will collide with that wall after time dt.

$\Rightarrow$ total number of particles colliding with wall per unit time

$$d\nu = \underbrace{\left(\frac{N}{V}\right)}_{=n} \underbrace{v_x}_{=\frac{p_x}{m}} \frac{e^{-p_x^2/2mkT} \int_{-\infty}^{\infty} dp_y\, dp_z\, e^{-\frac{p_y^2+p_z^2}{2mkT}}}{(2\pi mkT)^{3/2}}$$

$$\Rightarrow \nu = \frac{n}{m} \frac{\int_0^{\infty} dp_x\, p_x e^{-p_x^2/2mkT}}{(2\pi mkT)^{1/2}}$$

$$= \frac{n}{2m}(2mkT) \int_0^{\infty} \frac{du\, e^{-u}}{(2\pi mkT)^{1/2}} = \frac{n}{m}\sqrt{\frac{mkT}{2\pi}} =$$

$$= \frac{1}{4} n \sqrt{\frac{16kT}{2\pi m}} = \frac{1}{4} n \cdot \bar{v} = \nu$$ (= # particles hitting unit area per unit time)

here $\bar{v} = \sqrt{\frac{8kT}{\pi m}} = \langle |\vec{v}| \rangle$

is the average speed of 3d Maxwell's gas at temperature T.

These particles carry energy

$$E = \frac{n}{m} \frac{1}{(2\pi m k T)^{3/2}} \int_0^\infty dp_x \, p_x \, e^{-\frac{p_x^2}{2mkT}} \int dp_y \, dp_z \, e^{-\frac{p_y^2+p_z^2}{2mkT}} \cdot \frac{p_x^2+p_y^2+p_z^2}{2m}$$

$$= \frac{n}{2m^2 (2\pi m k T)^{3/2}} \int_0^{2\pi} d\varphi \int_0^{\pi/2} d\theta \, \sin\theta \, \underbrace{\cos\theta}_{\text{from } p_x} \int_0^\infty dp \, p^3 \cdot p^2 \, e^{-p^2/2mkT}$$

$$2\pi \int_0^1 d(\sin\theta) \sin\theta = \frac{1}{2} \int_0^\infty \frac{1}{2} d(p^2) \, (p^2)^2 e^{-\frac{p^2}{2mkT}}$$

$$= \frac{n}{2m^2 (2\pi m k T)^{3/2}} \left(2\pi \cdot \frac{1}{2} \cdot \frac{1}{2}\right) (2mkT)^3 \underbrace{\int_0^\infty du \, u^2 e^{-u}}_{2}$$

$$u = \frac{p^2}{2mkT}$$

$$= \frac{n\pi}{2m^2} \frac{(2mkT)^{3/2}}{\pi^{3/2}} = \frac{n}{2m^2} (2mkT) \sqrt{\frac{2mkT}{\pi}}$$

$$= n \sqrt{\frac{2kT}{\pi m}} (kT) = \cancel{\tfrac{1}{4}} (2kT) \underbrace{n \sqrt{\frac{kT}{2\pi m}}}_{\frac{1}{4} n \bar{v}} = (2kT) \nu$$

$\Rightarrow$ average energy that is carried out through the hole in the wall is

$$\mathcal{E} = \frac{E}{\nu} = 2kT > \frac{3}{2} kT \text{ inside the container!}$$